

①

Ex: $\int_0^{\infty} e^{-\sqrt{x}} dx$

Note

$$e^{-\sqrt{x}} \leq \frac{C}{x^2}, \quad x \geq a \quad (*)$$

Prove (*)

Put $x = t^2, \quad t > 0$

Want $e^{-t} \leq \frac{C}{t^4} \quad t \text{ large}$

$$t^4 e^{-t} \leq C \quad t \text{ large}$$

Let $F(t) = t^4 e^{-t}$

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Show $F(t) \leq C$ t large

$$\begin{aligned} F'(t) &= 4t^3 e^{-t} - t^4 e^{-t} \\ &= t^3 e^{-t} (4 - t) \end{aligned}$$

$$\boxed{\begin{aligned} F(t) &\leq F(4) \\ \text{all } t &\geq 0 \end{aligned}}$$

$$F'(t) \geq 0 \quad t \leq 4$$

$$F'(t) \leq 0 \quad t \geq 4$$

$$\begin{aligned} F(t) &\leq F(4) = 4^4 e^{-4} \\ &= 256 e^{-4} \end{aligned} \quad \text{all } t \geq 0$$

$$e^{-t} \leq \frac{256 e^{-4}}{t^4}, \quad t \geq 4$$

③

$$e^{-\sqrt{x}} \leq \frac{256e^{-4}}{x^2}, \quad x \geq 16$$

$$\int_{16}^{\infty} \frac{dx}{x^2} \quad \text{Conv}$$

$$\int_0^{\infty} e^{-\sqrt{x}} dx \quad \text{Conv.}$$

④

Limit Comparison Test

Assume f, g are bounded and have at most a finite # of disc. on $[a, t]$ for each $t > a$. Assume $f(x), g(x) > 0$ for $x > a$.

1. Assume $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L$ finite
($L=0$ is ok)

If $\int_a^{\infty} g(x) dx$ converges, so does
 $\int_a^{\infty} f(x) dx$

2. If $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L > 0$ (5) $(L = \infty \text{ is allowed})$

If $\int_a^{\infty} g(x) dx$ diverges then so

does $\int_a^{\infty} f(x) dx$

Ex: $\int_0^{\infty} \frac{dx}{\sqrt{1+x^4}}$

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$$\frac{1}{\sqrt{1+x^4}} \sim \frac{1}{x^2} \quad x \text{ large}$$

$$f(x) = \frac{1}{\sqrt{1+x^4}} \quad g(x) = \frac{1}{x^2}$$

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{x^2}{\sqrt{1+x^4}} = 1$$

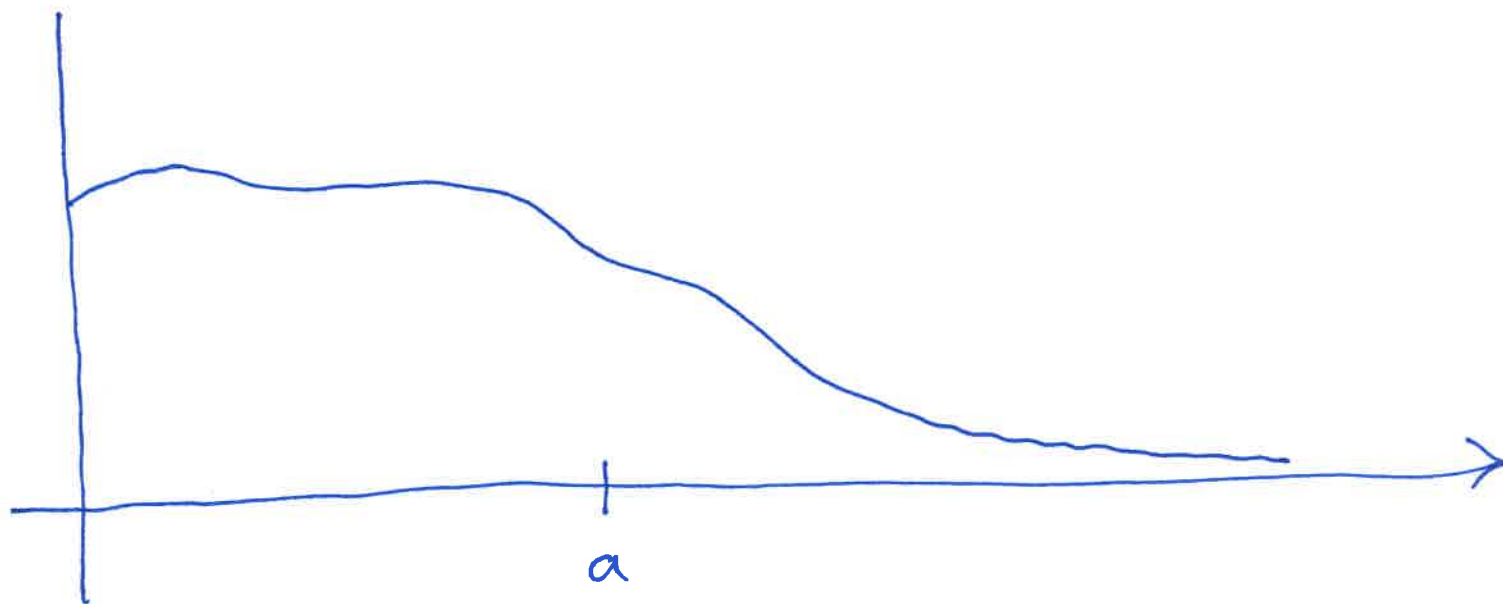
$$\int_1^{\infty} \frac{dx}{x^2} \text{ conv} \rightarrow \int_1^{\infty} \frac{dx}{\sqrt{1+x^4}} \text{ conv}$$

$$\boxed{\frac{1}{\sqrt{1+x^4}} \leq 1 \quad 0 \leq x \leq 1}$$

$$\rightarrow \int_0^{\infty} \frac{dx}{\sqrt{1+x^4}} \text{ conv.}$$

⑦

f bdd on $[0, \infty)$



$$\int_0^{\infty} f \text{ conv} \iff \int_a^{\infty} f \text{ conv.}$$

⑧

Functions of Several Variables

$\mathbb{R}^n$ - set of all n -tuples of real numbers

$$(x_1, x_2, x_3, \dots, x_n)$$

(x, y) or (t, x) when $n=2$

(x, y, z) or (t, x, y) when $n=3$

$$\vec{x} = (x_1, x_2, \dots, x_n)$$

$$\vec{y} = (y_1, y_2, \dots, y_n)$$

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$$\vec{x} + \vec{y} = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$$

$$c\vec{x} = (cx_1, cx_2, \dots, cx_n)$$

Dot Product or Inner Product

$$\vec{x} \cdot \vec{y} = x_1 y_1 + x_2 y_2 + \dots + x_n y_n$$

Notice $\vec{x} \cdot \vec{x} > 0$ unless $\vec{x} = \vec{0} = (0, 0, \dots, 0)$

Euclidean Norm, Length, or Magnitude

$$|\vec{x}| = \sqrt{\vec{x} \cdot \vec{x}} = \sqrt{\sum_{i=1}^n x_i^2}$$

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Cauchy - Schwarz Inequality

$$|\vec{x} \cdot \vec{y}| \leq |\vec{x}| |\vec{y}|$$

← Product of Length x and Length y

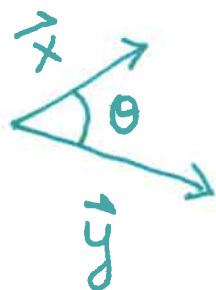
abs value of $\vec{x} \cdot \vec{y}$

In 2 or 3 dimensions

$$\vec{x} \cdot \vec{y} = |\vec{x}| |\vec{y}| \cos \theta$$

θ = angle between $\vec{x}, \vec{y}$

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2 or 3 Dim

Cauchy-Schwarz says

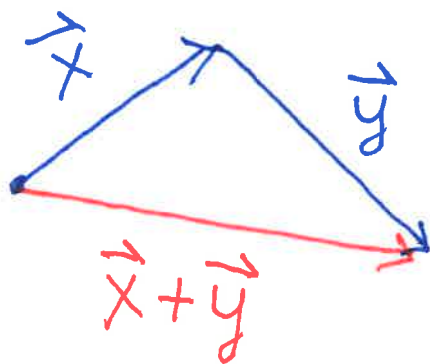
$$|\cos \theta| \leq 1$$

Could Use This to define
angles between vectors in
higher Dimensions

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Triangle Inequality

$$|\vec{x} + \vec{y}| \leq |\vec{x}| + |\vec{y}|$$



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Fns with real number for input
Vector for Output

$$\vec{F}(t) = (F_1(t), F_2(t), \dots, F_n(t))$$

Derivative $\vec{F}'(t) = (F_1'(t), F_2'(t), \dots, F_n'(t))$

Integral $\int_a^b \vec{F}(t) dt = \left(\int_a^b F_1(t) dt, \dots, \int_a^b F_n(t) dt \right)$

Mean Value Theorem

$$\vec{F}'(c) = \frac{\vec{F}(b) - \vec{F}(a)}{b-a}$$

NOT VALID

$$F_1'(c_1) = \frac{F_1(b) - F_1(a)}{b-a}$$

$$F_2'(c_2) = \frac{F_2(b) - F_2(a)}{b-a}$$

$$\vdots$$
The c_i 's might
be different

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Ex: $\vec{F}(t) = (\cos t, \sin t)$

$$a=0$$

$$b=2\pi$$

$$\vec{F}'(t) = (-\sin t, \cos t)$$

$$\frac{\vec{F}(b) - \vec{F}(a)}{b-a} = \frac{(1,0) - (1,0)}{2\pi} = (0,0)$$

$$(-\sin(c), \cos(c)) = (0,0)$$

Impossible

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Real-Valued Fns of Several Variables

Vector Input $\rightarrow$ Scalar Output

Look at 3 variables first

$$\Omega \subset \mathbb{R}^3$$

$$\vec{X} = (x_1, x_2, x_3)$$

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Partial Derivatives

$$f_{x_1}(x_1, x_2, x_3) = \lim_{h \rightarrow 0} \frac{f(x_1+h, x_2, x_3) - f(x_1, x_2, x_3)}{h}$$

$$f_{x_2}(x_1, x_2, x_3) = \lim_{h \rightarrow 0} \frac{f(x_1, x_2+h, x_3) - f(x_1, x_2, x_3)}{h}$$

Similarly for f_{x_3}

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Common to write

$\frac{\partial f}{\partial x_i}$ in place of f_{x_i}

$$\frac{\partial}{\partial x_2} (x_1 e^{x_2^2} \sin x_3) = x_1 (e^{x_2^2} \cdot 2x_2) \sin x_3$$

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Ex: $f(x,y) = x^3 y^2 + e^{xy}$

$$f_x(x,y) = 3x^2 y^2 + e^{xy}(y)$$

$$f_y(x,y) = 2x^3 y + e^{xy}(x)$$

Ex: $f(x_1, x_2, x_3, x_4) = x_4^{10} + \sqrt{1+x_2^2} e^{x_3} + \sin(2x_1)$

$$f_{x_2} = e^{x_3} \frac{x_2}{\sqrt{1+x_2^2}}$$

$$f_{x_1} = 2 \cos(2x_1)$$

Pitfall of Notation

$$f(x, y) = x^2 + y^3 e^x$$

What does $f_y(y, x)$

or $\frac{\partial}{\partial y} f(y, x)$ mean?

Could mean Take $\frac{\partial}{\partial y} f(x, y) = 3y^2 e^x$

Then switch x, y

or $f(y, x) = y^2 + x^3 e^y$ Then diff

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Another Notation

$$f(s, t) = s^2 + t^3 e^s$$

Instead of $f_s, f_t, \frac{\partial f}{\partial s}, \frac{\partial f}{\partial t}$

$f_{,1}(s, t) = \text{part. deriv wrt 1st arg}$

$f_{,2}(s, t) = \text{part. deriv wrt 2nd arg}$

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$$f_{,1}(s,t) = 2s + t^3 e^s$$

$$f_{,2}(s,t) = 3t^2 e^s$$

$$f_{,1}(s,s) = 2s + s^3 e^s$$

$$f_{,2}(s^3, t^2) = 3t^4 e^{s^3}$$

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Gradient of $f(\vec{x})$

$$f(\vec{x}) = f(x_1, x_2, \dots, x_n)$$

$$\begin{aligned}\vec{\nabla} f(\vec{x}) &= (f_{,1}(\vec{x}), f_{,2}(\vec{x}), \dots, f_{,n}(\vec{x})) \\ &= \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right)\end{aligned}$$

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Ex: $n=3$ $f(\vec{x}) = x_1^2 x_2^3 + x_3^8$

$$f_{,1}(\vec{x}) = 2x_1 x_2^3$$

$$f_{,2}(\vec{x}) = 3x_1^2 x_2^2$$

$$f_{,3}(\vec{x}) = 8x_3^7$$

$$\vec{\nabla} f(\vec{x}) = (2x_1 x_2^3, 3x_1^2 x_2^2, 8x_3^7)$$

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General Linear Appx

for $f(\vec{x})$ near $\vec{x}^*$

$$n=1 \quad f(x) \approx f(x^*) + f'(x^*)(x - x^*)$$

$$f(\vec{x}) \approx f(\vec{x}^*) + \vec{\nabla} f(\vec{x}^*) \cdot (\vec{x} - \vec{x}^*)$$

$$f(\vec{x}^*) + \sum_{i=1}^n \frac{\partial f}{\partial x_i}(\vec{x}^*) (x_i - x_i^*)$$

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Higher Derivatives

$$f_{x_i x_j} = \frac{\partial}{\partial x_j} \left(\frac{\partial f}{\partial x_i} \right) = \frac{\partial^2 f}{\partial x_j \partial x_i}$$

Thm: Assume all 2nd order partial derivs are continuous. The

$$f_{x_i x_j} = f_{x_j x_i}$$

Order DOES NOT matter

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$$f(x, y) = x^3 y^5 + (\sin x) e^{3y}$$

$$f_x(x, y) = 3x^2 y^5 + (\cos x) e^{3y}$$

$$f_{xy}(x, y) = 15x^2 y^4 + 3(\cos x) e^{3y}$$

$$f_y(x, y) = 5x^3 y^4 + 3(\sin x) e^{3y}$$

$$f_{yx}(x, y) = 15x^2 y^4 + 3(\cos x) e^{3y}$$

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$$f(x,y) = e^{x^3} \ln(1+x^2) y^2$$

$$f_{xxxxxxxyxyxxxxxy} = 0$$

Do the y -derivs 1st

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Rmk: $f_{xxxyyxxyy} = f_{xyxyxyxy}$
 $= f_{xxxxyyyy}$
 $= f_{yyyyxxxx}$
 $= f_{xyyyxyxx}$

assuming 8th order
partials are cont.

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Hessian Matrix

Hessian Matrix for $f(\vec{x})$ is

$$H_f(\vec{x}) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_1 \partial x_3} & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \dots & \dots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \dots & \dots & \dots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}$$